

SIEMENS

SINUMERIK

SINUMERIK ONE Template Machines Create & Run MyVirtual Machine

Installation Manual

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Valid for:
MACHINUM
Create & Run MyVirtual Machine from V1.4
SINUMERIK ONE
SINUMERIK Virtual CNC software from V6.23

01/2024

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Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury will result if proper precautions are not taken.
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WARNING
--

indicates that death or severe personal injury may result if proper precautions are not taken.

CAUTION
--

indicates that minor personal injury can result if proper precautions are not taken.
--

NOTICE

indicates that property damage can result if proper precautions are not taken.
--

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING
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Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.
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Trademarks

All names identified by ® are registered trademarks of Siemens Aktiengesellschaft. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Introduction

1.1 About SINUMERIK

From simple, standardized CNC machines to premium modular machine designs – the SINUMERIK CNCs offer the right solution for all machine concepts. Whether for individual parts or mass production, simple or complex workpieces – SINUMERIK is the highly dynamic automation solution, integrated for all areas of production. From prototype construction and tool design to mold making, all the way to large-scale series production.

Visit our website for more information SINUMERIK (<https://www.siemens.com/sinumerik>).

1.2 About this documentation

Target group

This documentation is intended for project engineers, programmers and commissioning engineers.

Purpose

This documentation describes the installation environment required for working with machine project templates for Create & Run MyVirtual Machine.

The machine projects are available via installation package "Template Machines for Create/Run MyVirtual Machine CNC-SW X.XX", which is compatible to a SINUMERIK Virtual CNC-SW. The package includes machine projects for different machine tool types, which serve as machine example in Run MyVirtual Machine and as a template for machine projects in Create MyVirtual Machine. The installation package can only be used in conjunction with a virtual SINUMERIK environment.

Benefits

This documentation brings the addressed target group into a position to be able to install the machine projects to use these in Create or run MyVirtual Machine.

Standard scope

This documentation only describes the functionality of the standard version. This may differ from the scope of the functionality of the system that is actually supplied. Please refer to the ordering documentation only for the functionality of the supplied drive system.

It may be possible to execute other functions in the system which are not described in this documentation. This does not, however, represent an obligation to supply such functions with a new control or when servicing.

For reasons of clarity, this documentation cannot include all of the detailed information on all product types. Further, this documentation cannot take into consideration every conceivable type of installation, operation and service/maintenance.

The machine manufacturer must document any additions or modifications they make to the product themselves.

Websites of third-party companies

This document may contain hyperlinks to third-party websites. Siemens is not responsible for and shall not be liable for these websites and their content. Siemens has no control over the information which appears on these websites and is not responsible for the content and information provided there. The user bears the risk for their use.

1.3 Documentation on the internet

1.3.1 Documentation overview SINUMERIK ONE

Comprehensive documentation about the functions provided in SINUMERIK ONE Version 6.13 and higher is provided in the Documentation overview SINUMERIK ONE (<https://support.industry.siemens.com/cs/ww/en/view/109768483>).



You can display documents or download them in PDF and HTML5 format.

The documentation is divided into the following categories:

- User: Operating
- User: Programming
- Manufacturer/Service: Functions
- Manufacturer/Service: Hardware
- Manufacturer/Service: Configuration/Setup
- Manufacturer/Service: Safety Integrated

- Information and training
- Manufacturer/Service: SINAMICS

1.3.2 Documentation overview SINUMERIK operator components

Comprehensive documentation about the SINUMERIK operator components is provided in the Documentation overview SINUMERIK operator components (<https://support.industry.siemens.com/cs/document/109783841/technische-dokumentation-zu-sinumerik-bedienkomponenten?dti=0&lc=en-WW>).

You can display documents or download them in PDF and HTML5 format.

The documentation is divided into the following categories:

- Operator Panels
- Machine control panels
- Machine Pushbutton Panel
- Handheld Unit/Mini handheld devices
- Further operator components

An overview of the most important documents, entries and links to SINUMERIK is provided at SINUMERIK Overview - Topic Page (<https://support.industry.siemens.com/cs/document/109766201/sinumerik-an-overview-of-the-most-important-documents-and-links?dti=0&lc=en-WW>).

1.4 Feedback on the technical documentation

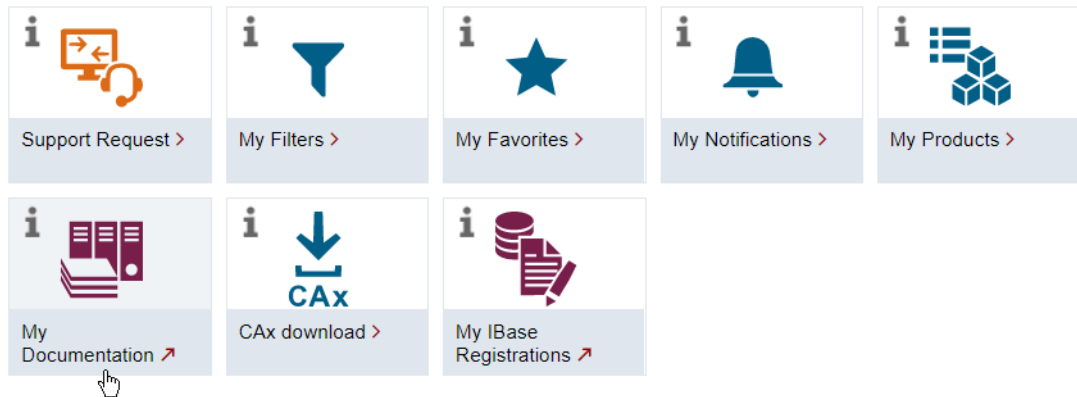
If you have any questions, suggestions, or corrections regarding the technical documentation published in the Siemens Industry Online Support, use the link "Give feedback" link which appears at the end of the entry.

1.5 mySupport documentation

With the "mySupport documentation" web-based system you can compile your own individual documentation based on Siemens content, and adapt it for your own machine documentation.

To start the application, click on the "My Documentation" tile on the "mySupport links and tools" (<https://support.industry.siemens.com/cs/ww/en/my>) portal page:

mySupport Links and Tools



The configured manual can be exported in RTF, PDF or XML format.

Note

Siemens content that supports the mySupport documentation application can be identified by the presence of the "Configure" link.

1.6 Service and Support

Product support

You can find more information about products on the internet:

Product support (<https://support.industry.siemens.com/cs/ww/en/>)

The following is provided at this address:

- Up-to-date product information (product announcements)
- FAQs (frequently asked questions)
- Manuals
- Downloads
- Newsletters with the latest information about your products
- Global forum for information and best practice sharing between users and specialists
- Local contact persons via our Contacts at Siemens database (→ "Contact")
- Information about field services, repairs, spare parts, and much more (→ "Field Service")

Technical support

Country-specific telephone numbers for technical support are provided on the internet at address (<https://support.industry.siemens.com/cs/ww/en/sc/4868>) in the "Contact" area.

If you have any technical questions, please use the online form in the "Support Request" area.

Training

You can find information on SITRAIN at the following address (<https://www.siemens.com/sitrain>).

SITRAIN offers training courses for automation and drives products, systems and solutions from Siemens.

Siemens support on the go



With the award-winning "Industry Online Support" app, you can access more than 300,000 documents for Siemens Industry products – any time and from anywhere. The app can support you in areas including:

- Resolving problems when implementing a project
- Troubleshooting when faults develop
- Expanding a system or planning a new system

Furthermore, you have access to the Technical Forum and other articles from our experts:

- FAQs
- Application examples
- Manuals
- Certificates
- Product announcements and much more

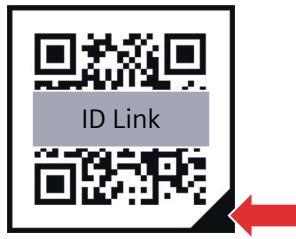
The "Industry Online Support" app is available for Apple iOS and Android.

Digital product information as QR code

The ID link according to IEC 61406 provides you with digital information about your product.

The ID link is a globally unique identifier and appears as a QR code on the product.

You can recognize the ID link by the frame with a black corner at the bottom right.



In addition to the digital nameplate you will find the following product information:

- Technical specifications
- FAQs
- Manuals
- Certificates
- Product announcements
- Application examples

1.7 Using OpenSSL

This product can contain the following software:

- Software developed by the OpenSSL project for use in the OpenSSL toolkit
- Cryptographic software created by Eric Young.
- Software developed by Eric Young

You can find more information on the internet:

- OpenSSL (<https://www.openssl.org>)
- Cryptsoft (<https://www.cryptsoft.com>)

1.8 Compliance with the General Data Protection Regulation

Siemens observes standard data protection principles, in particular the data minimization rules (privacy by design).

For this product, this means:

The product does not process or store any personal data, only technical function data (e.g. time stamps). If the user links this data with other data (e.g. shift plans) or if he/she stores person-related data on the same data medium (e.g. hard disk), thus personalizing this data, he/she must ensure compliance with the applicable data protection stipulations.

Fundamental safety instructions

2.1 General safety instructions

 WARNING
Danger to life if the safety instructions and residual risks are not observed If the safety instructions and residual risks in the associated hardware documentation are not observed, accidents involving severe injuries or death can occur. • Observe the safety instructions given in the hardware documentation. • Consider the residual risks for the risk evaluation.

 WARNING
Malfunctions of the machine as a result of incorrect or changed parameter settings As a result of incorrect or changed parameterization, machines can malfunction, which in turn can lead to injuries or death. • Protect the parameterization against unauthorized access. • Handle possible malfunctions by taking suitable measures, e.g. emergency stop or emergency off.

2.2 Warranty and liability for application examples

Application examples are not binding and do not claim to be complete regarding configuration, equipment or any eventuality which may arise. Application examples do not represent specific customer solutions, but are only intended to provide support for typical tasks.

As the user you yourself are responsible for ensuring that the products described are operated correctly. Application examples do not relieve you of your responsibility for safe handling when using, installing, operating and maintaining the equipment.

2.3 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

2.3 Cybersecurity information

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit
<https://www.siemens.com/cybersecurity-industry>.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under
<https://new.siemens.com/cert>.

Further information is provided on the Internet:

Industrial Security Configuration Manual (<https://support.industry.siemens.com/cs/ww/en/view/108862708>)



WARNING

Unsafe operating states resulting from software manipulation

Software manipulations, e.g. viruses, Trojans, or worms, can cause unsafe operating states in your system that may lead to death, serious injury, and property damage.

- Keep the software up to date.
- Incorporate the automation and drive components into a state-of-the-art, integrated industrial cybersecurity concept for the installation or machine.
- Make sure that you include all installed products in the integrated industrial cybersecurity concept.
- Protect files stored on exchangeable storage media from malicious software by with suitable protection measures, e.g. virus scanners.
- Carefully check all cybersecurity-related settings once commissioning has been completed.

Installation environment

3.1 General information

Template machine projects

The software package includes machine projects ("*.vcp"), which are used as a template project or as a sample project for Create & Run MyVirtual Machine.

Note

The machine projects (Template Machines) are only compatible with the corresponding version of the SINUMERIK Virtual CNC software. If you are using several versions of SINUMERIK Virtual CNC software, then you must install the matching machine projects; for example SINUMERIK Virtual CNC-SW V6.20 with Template Machines V6.20.

Sample machine projects are installed for the following machine tool types:

- Machines for use in the Run MyVirtual Machine demo mode
- 3 axis milling machine
- 3+2 axis milling machine
- 5 axis milling machine, multitasking
- Basic turning machine
- Basic turning machine with driven tools
- Robotics machines

Every machine project has the following scope:

- CNC machine that has been configured and commissioned
- Examples of NC programs that can be run; all of the necessary tools are already defined in the tool list
- Matching 3D machine model for 3D simulation
- Libraries with examples for blanks, clamping arrangements (protection areas) and configurations

Using machine projects license-free in Run MyVirtual Machine demo mode (as of CNC software V6.22)

The installation scope of the machine projects includes machines that you can use without a license in Run MyVirtual Machine demo mode. In this variant you can only open and work with special marked machine projects. Demo machines do not support the full scope of functions of Run MyVirtual Machine or the SINUMERIK CNC software. Therefore, the changes (e.g. modified NC programs, created clamping, etc.) to the started projects cannot be saved. The projects are reset to the initial state each time the machine project is restarted.

3.1 General information

The following machines are available as template machines in Run MyVirtual Machine demo mode. The demo machine projects are marked with the addition (Demo) in the machine project name.

- Turning machine (SinuTurn-DT (Demo).vcp)
1-channel, 3-axis machine with main spindle and driven tool spindle
- Milling machine (SinuMill3 (Demo).vcp)
1-channel, 3-axis machine
- Milling machine (SinuMill3+2-AC (Demo).vcp)
1-channel, 3+2-axis machine
- Turning machine with robot (Robotics6-Handling (Demo).vcp)
Turning machine with main spindle and handling robot

Installation environment overview

The installation packet installs machine projects in MyVirtual Machine products. The software package is only functional in conjunction with additional MyVirtual Machine products.

3.2 Installation guideline

Installing and configuring Create & Run MyVirtual Machine and additional software

The virtual development and configuration environment for SINUMERIK ONE consists of several individual products which are installed and configured prior to their first use. All necessary work steps for the application in question are shown in the overview. In the right-hand column of the overview you will find the reference to further information.

3.2 Installation guideline

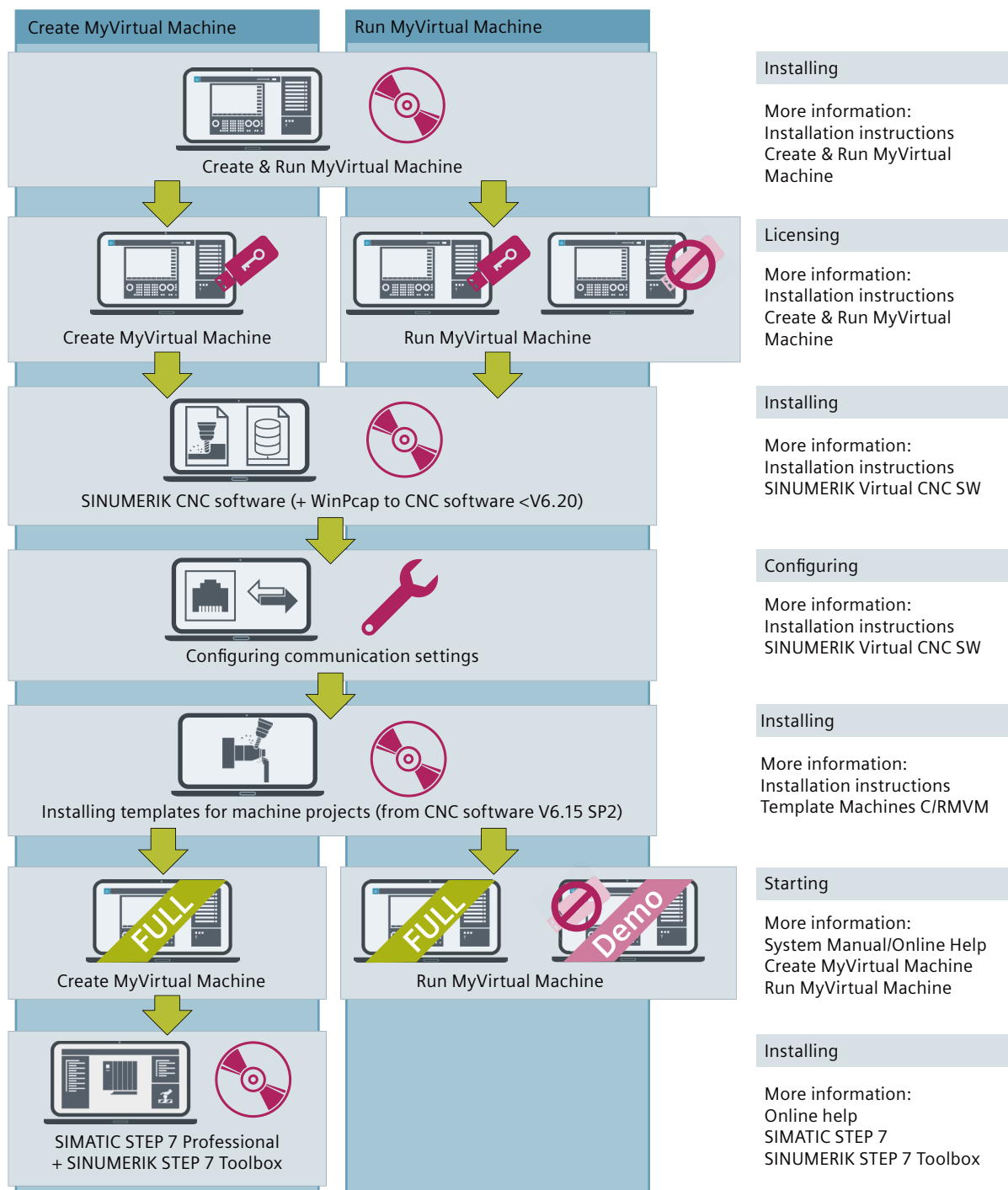


Figure 3-1 Guideline: Installing and configuring MyVirtual Machine products (sample installation sequence)
After you have completed all steps, Create & Run MyVirtual Machine is ready to use.

System requirements

4.1 System requirements

Installation requirement

You require administrator rights to install templates for machine projects. After installation, use a user account without administrator rights in order to protect the system against security risks and unauthorized manipulation.

See also: Cybersecurity information (Page 11)

System requirements

Installed software

Templates for machine projects can only be installed and licensed if the following software has already been installed.

- SINUMERIK Virtual CNC SW
- Create & Run MyVirtual Machine, compatible with the respective SINUMERIK Virtual CNC SW version

System requirements and compatibility

The requirements for installing the SINUMERIK Virtual CNC software and Create & Run MyVirtual Machine apply. For more information, e.g., as to which version of Create & Run MyVirtual Machine is compatible with the SINUMERIK Virtual CNC software version, refer to the installation instructions of the corresponding software products.

4.1 System requirements

Installing

5.1 Installing machine projects

Description

Installation package "Template Machines for Create/Run MyVirtual Machine CNC-SW X.XX" is provided on the installation medium supplied.

Note

The term "installation medium" is synonymous with DVD, download or other supplied media.

Requirement

- The hardware and software of the PC meet the system requirements.
- You have administrator rights on your computer.
- All running programs are closed.
- The appropriate version of SINUMERIK Virtual CNC-SW has already been installed

This is how you install machine projects

1. On the installation medium navigate to the set-up files.
2. Double-click "Start.exe". The installation wizard opens.
3. Follow the wizard's instructions and confirm the prompts.

The machine projects are installed and can be used with the appropriate, additional software products, e.g. Create MyVirtual Machine.

5.2 Uninstalling machine projects

Overview

Templates for machine projects that have been installed cannot be separately uninstalled. The templates are automatically uninstalled if the corresponding version of Virtual SINUMERIK CNC software is uninstalled.

More information

More information is provided in the *Installing SINUMERIK Virtual CNC Software Installation Manual*.

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